

Chapter 2 Basic concepts from probability theory

XIE Wenxian

School of mathematics and statistics
Northwestern Polytechnical University

September 24, 2024

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Objective

- In this chapter, we will review some results from probability theory that are essential for the development of the statistical results.
- Andrei Kolmogorov (1903-87) laid the mathematical foundations of probability theory and the theory of randomness.
 - Foundations of Theory of Probability.
 - He developed the probability theory from an axiomatic point of view.
- Probability theory provides a mathematical model for the study of randomness and uncertainty.

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Random experiment

- Experiment: Any process whose outcome is not known in advance but is random.
- Trial: Assume that the experiment can be repeated any number of times under identical conditions, then each repetition is called a trial.

A random experiment satisfies the following three conditions:

- The set of all possible outcomes is known in advance in each trial;
- In any particular trial, it is not known which particular outcome will happen;
- The experiment can be repeated under identical conditions.

Some basic definitions

- Sample space

- The set consisting of all possible outcomes
- It is referred to a probability space
- S

- Sample point

- An outcome in S

- Event

- $A \subset S$
- impossible event ϕ

- Set operation

- $A \cap B, A \cup B, A^c$
- Definition 2.2.1 Mutually exclusive (disjoint) events: $A \cap B = \phi$

Four definitions of probability

- Informal definition of probability
- Classical definition of probability
- Frequency definition of probability
- Axiomatic definition of probability

Definition 2.2.2: The probability of an event is a measure (number) of the chance with which we can expect the event to occur. We assign a number between 0 and 1 inclusive to the probability of an event.

- A probability of 1 means that we are 100% sure of the occurrence of an event;
- A a probability of 0 means that we are 100% sure of the nonoccurrence of the event.
- The probability of any event A in the sample space S is denoted by $P(A)$.
- $P(S) = 1$

Classical definition of probability

Definition 2.2.3: If there are n equally likely possibilities, of which one must occur, and m of these are regarded as favorable to an event, or as “success,” then the probability of the event or a “success” is given by $\frac{m}{n}$.

Method of computing probability when all outcomes are equally likely:

- Count the number of outcomes in the sample space; say this is n .
- Count the number of outcomes in the event of interest, A , and say this is m .
- $P(A) = \frac{m}{n}$

Classical definition of probability

Method of computing probability when all outcomes are not equally likely:

- Let $\{\omega_i\}, i = 1 \dots n$ be the outcomes of the sample space S . Let $P(\{\omega_i\}) = p_i, i = 1 \dots n$. In this case, the probability of each outcome, p_i , is assumed to be known.
- List all the outcomes in the event A , say, $\omega_i, \omega_j \dots \omega_m$.
- $P(A) = P(\{\omega_i\}) + P(\{\omega_j\}) + \dots + P(\{\omega_m\}) = p_i + p_j + \dots + p_m$, the sum of the probabilities of the outcomes in A .

Example 2.2.1. A balanced die (with all outcomes equally likely) is rolled. Let A be the event that an even number occurs. Find $P(A)$.

Solution there are three favorable outcomes (2, 4, 6) in A , and the sample space has six elements, (1, 2, 3, 4, 5, 6). Hence, $P(A) = 3/6 = 1/2$.

Classical definition of probability

Example 2.2.2. Suppose we toss two coins. Assume that all the outcomes are equally likely (fair coins).

- (a) What is the sample space?
- (b) Let A be the event that at least one of the coins shows up heads. Find $P(A)$.
- (c) What will be the sample space if we know that at least one of the coins showed up heads?

Classical definition of probability

Solution

- (a) The sample space consists of four outcomes, namely $S = \{(H, H), (H, T), (T, H), (T, T)\}$.
- (b) The event A has three outcomes, (H, H) , (H, T) , and (T, H) . Therefore $P(A) = 3/4$.
- (c) Since we know that at least one of the coins showed up heads, the possible outcomes are (H, H) , (H, T) , and (T, H) . The sample space now has only three outcomes $\{(H, H), (H, T), (T, H)\}$.

The frequency interpretation of probability

Definition 2.2.4: The probability of an outcome (event) is the proportion of times the outcome (event) would occur in a long run of repeated experiments.

- Example: to find the probability of heads, H , using a biased coin, we would imagine the coin is repeatedly tossed. Let $n(H)$ be the number of times H appears in n trials. Then the probability of heads is defined as

$$P(H) = \lim_{n \rightarrow \infty} \frac{n(H)}{n}.$$

Axiomatic definition of probability

Definition 2.2.5: Let S be a sample space of an experiment. Probability $P(\cdot)$ is a real-valued function that assigns to each event A in the sample space S a number $P(A)$, called the probability of A , with the following conditions satisfied:

- It is nonnegative, $P(A) \geq 0$.
- It is unity for a certain event. That is, $P(S) = 1$.
- It is additive over the union of an infinite number of pairwise disjoint events, that is, if $A_1, A_2, \dots$ form a sequence of pairwise mutually exclusive events (that is, $A_i \cap A_j = \phi$, for $i \neq j$) in S , then
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Some basic properties of probability

For two events A and B in S , we have the following:

- $P(A^c) = 1 - P(A)$
- If $A \subset B$, then $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. In particular, if $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$

Example 2.2.3 A die is loaded (not all outcomes are equally likely) such that the probability that the number i shows up is $K \cdot i, i = 1, 2, \dots, 6$, where K is a constant. Find

- (a) the value of K .
- (b) the probability that a number greater than 3 shows up.

Some basic properties of probability

Solution

- (a) Here the sample space S has six outcomes 1, 2, .., 6. Hence, using axioms (2) and (3) we have

$$P(\{1\}) + P(\{2\}) + \cdots + P(\{6\}) = 1.$$

Since $P(i) = Ki$, we have

$$K \cdot 1 + K \cdot 2 + \cdots + K \cdot 6 = 1.$$

Hence $K = 1/21$.

- (b) Let A be the event that a number greater than 3 shows up. Then the outcomes in A are 4, 5, 6 and they are mutually exclusive. Therefore,

$$\begin{aligned} P(A) &= P(\{4\}) + P(\{5\}) + P(\{6\}) \\ &= \frac{4}{21} + \frac{5}{21} + \frac{6}{21} = \frac{15}{21}. \end{aligned}$$

Some basic properties of probability

Example 2.2.4 In a large university, the freshman profile for 1 year's fall admission says that 40% of the students were in the top 10% of their high school class, and that 65% are white, of whom 25% were in the top 10% of their high school class. What is the probability that a freshman student selected randomly from this class either was in the top 10% of his or her high school class or is white?

Solution Let E_1 be the event that a person chosen at random was in the top 10% of his or her high school class, and let E_2 be the event that the student is white. We are given $P(E_1) = 0.40$, $P(E_2) = 0.65$, and $P(E_1 \cap E_2) = 0.25$. Then the event that the student chosen is white or was in the top 10% of his or her high school class is represented by $E_1 \cup E_2$. Thus

$$\begin{aligned} P(E_1 \cup E_2) &= P(E_1) + P(E_2) - P(E_1 \cap E_2) \\ &= 0.4 + 0.65 - 0.25 = 0.80. \end{aligned}$$

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities**
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Multiplication principle

Theorem 2.3.1 If the experiments $A_1, A_2, \dots, A_m$ contain, respectively, $n_1, n_2, \dots, n_m$ outcomes, such that for each possible outcome of A_1 there are n_2 possible outcomes for A_2 , and so on, then there are a total of $n_1 n_2 \cdots n_m$ possible outcomes for the composite experiment $A_1, A_2, \dots, A_m$.

Example 2.3.1 In how many different ways can a student club at a large university with 500 members choose its president and vice president?

Solution The president can be chosen 500 ways, and the vice president can be chosen from the remaining 499 ways. Hence, by the multiplication principle, there are $(500)(499) = 249,500$ ways in which the complete choice can be made.

Four sampling methods

(1) Sampling with Replacement and the Objects Are Ordered

- When a random sample of size k is taken with replacement from a total of n objects and the objects being ordered, then there are n^k possible ways of selecting k -tuples.

(2) Sampling without Replacement and the Objects Are Ordered

- **Permutation**: any particular (ordered) arrangement of these objects if r objects are chosen from a set of n distinct objects without replacement.
- $n! = n(n-1) \cdots (2)(1)$
- The number of **permutations of m objects selected from a collection of n distinct objects** is ${}_nP_m = \frac{n!}{(n-m)!} = n(n-1) \cdots (n-m+1)$

Four sampling methods

Example 2.3.2 How many distinct three-digit numbers can be formed using the digits 2, 4, 6, and 8 if no digit can be repeated?

Solution The number of distinct three-digit numbers will be the number of permutations of three numbers from the set of four numbers 2, 4, 6, 8. Hence, the number of distinct three-digit numbers will be ${}_4P_3 = 4! / 1! = 24$.

(3) Sampling without Replacement and the Objects Are Not Ordered

- The number of ways in which m objects can be selected (without replacement) from a collection of n distinct objects is

$$\begin{aligned}\binom{n}{m} &= \frac{n!}{m!(n-m)!} \\ &= \frac{n(n-1)(n-2) \cdots (n-m+1)}{m!}, \quad m = 0, 1, 2, \dots, n.\end{aligned}$$

- $\binom{n}{m}$ is to be read as “ n choose m .”
- When the order of arrangement is not important, for example, if we do not distinguish between AB and BA , the arrangement is called a **combination**.

(4) Sampling with Replacement and the Objects Are Not Ordered

- In obtaining an unordered sample of size k , with replacement, from a total of n objects, $(k - 1)$ replacements will be made before sampling ceases.
- n is increased by $(k - 1)$ so that sampling in this manner may be thought of as drawing an unordered sample of size k from a population of size $(n + k - 1)$
- The number of possible samples can be obtained by

$$\binom{n + k - 1}{k} = \frac{(n + k - 1)!}{k! (n - 1)!} \quad k = 0, 1, 2, \dots$$

Four sampling methods

Example 2.3.4 An urn contains 15 balls numbered 1 to 15. If four balls are drawn at random, with replacement and without regard for order, how many samples are possible?

Solution Using the previous formula, the number of possible samples is

$$\binom{15 + 4 - 1}{4} = \frac{18!}{4! 14!} = 3060.$$

If we need to divide n objects into more than two groups, we can use the following result.

Number of combinations of N objects into M classes

The number of ways that n objects can be grouped into m classes with n_i in the i th class, $i = 1, 2, \dots, m$ and $\sum_{i=1}^m n_i = n$ is given by

$$\binom{n}{n_1 n_2 \dots n_m} = \frac{n!}{n_1! n_2! \dots n_m!} \text{ (multinomial coefficients)}$$

Example 2.3.6

In a tank containing 10 fish, there are three yellow and seven black fish. We select three fish at random.

- 1 What is the probability that exactly one yellow fish gets selected?
- 2 What is the probability that at most one yellow fish gets selected?
- 3 What is the probability that at least one yellow fish gets selected?

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Conditional probability

If we know that an event has already occurred or we have some partial information about the event, then this knowledge may affect the probability of the event of interest.

Definition 2.4.1. The conditional probability of an event A , given that an event B has occurred, denoted by $P(A|B)$, is equal to

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) > 0$$

- The conditional probability $P(\cdot|B)$ is a probability on event B .
- It satisfies all the axioms of a probability.

Conditional probability

Example 2.4.1 We toss two balanced dice, and let A be the event that the sum of the face values of two dice is 8, and B be the event that the face value of the first one is 3. Calculate $P(A|B)$.

Solution

The elements of the events A and B are

$$A = \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}.$$

and

$$B = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}.$$

$$\text{Now } A \cap B = \{(3, 5)\}$$

$$P(A) = 5/36, P(B) = 6/36, \text{ and } P(A \cap B) = 1/36.$$

Therefore,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{36}}{\frac{6}{36}} = \frac{1}{6}.$$

Some properties of conditional probability

- If $E_2 \subset E_1$, then $P(E_2 | A) \leq P(E_1 | A)$.
- $P(E | A) = 1 - P(E^C | A)$.
- $P(E_1 \cup E_2 | A) = P(E_1 | A) + P(E_2 | A) - P(E_1 \cap E_2 | A)$.
- Multiplication law: $P(A \cap B) = P(B)P(A | B) = P(A)P(B | A)$. In general,

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \dots \\ P(A_n | A_1 \cap A_2 \cap \dots \cap A_{n-1}).$$

Some properties of conditional probability

Example 2.4.2 A fruit basket contains 25 apples and oranges, of which 20 are apples. If two fruits are randomly picked in sequence, what is the probability that both the fruits are apples?

Solution

Let

$A = \{\text{event that the first fruit is an apple}\}$

$B = \{\text{event that the second fruit is an apple}\}.$

We need to find $P(A \cap B)$. We have

$$P(A) = 20/25, \quad P(B|A) = 19/24.$$

Now using the multiplication principle for conditional probabilities,

$$P(A \cap B) = P(A)P(B|A) = \left(\frac{20}{25}\right)\left(\frac{19}{24}\right) = 0.633.$$

Hence, the probability that both the fruits are apples is 0.633.

Independent

- **Definition 2.4.2** Two events A and B with $P(A) \neq 0$ and $P(B) \neq 0$ are said to be **independent** if $P(A|B) = P(A)$, or $P(B|A) = P(B)$. Otherwise, A and B are dependent.
- A and B are independent if $P(A \cap B) = P(A)P(B)$.
- The k events $A_1, A_2, \dots, A_k$ are mutually independent if for every $j = 2, 3, \dots, k$ and every subset of distinct indices $i_1, i_2, \dots, i_j$

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_j}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_j})$$

- Mutually independent events will often be called independent. In particular, if $P(A_{i_j} \cap A_{i_k}) = P(A_{i_j})P(A_{i_k})$ for each $j \neq k$, then the events are called **pairwise independent**.

Example 2.4.4 Suppose that we toss two fair dice. Let E_1 denote the event that the sum of the dice is 6 and E_2 denote the event that the first die equals 4. Then, $P(E_1 \cap E_2) = P\{4, 2\} = \frac{1}{36} \neq P(E_1)P(E_2) = \frac{5}{216}$. Hence, E_1 and E_2 are dependent events.

Law of total probability

Theorem 2.4.1 Assume $S = A_1 \cup A_2 \cup \dots \cup A_n$, where $P(A_i) > 0$, $i = 1, 2, \dots, n$, and $A_i \cap A_j = \phi$ (null set) for $i \neq j$. Then for any event B ,

$$P(B) = \sum_{i=1}^n P(A_i)P(B|A_i)$$

- The set $A_1, A_2, \dots, A_n$ given in the law of total probability is called the partition of S .

Example 2.4.5 Assume that a noisy channel independently transmits symbols, say 0s 60% of the time and 1s 40% of the time. At the receiver, there is a 1% chance of obtaining any particular symbol distorted. What is the probability of receiving a 1, irrespective of which symbol is transmitted?

Law of total probability

Solution

Given

$$P(0) = P(0' \text{ is transmitted}) = 0.6$$

and

$$P(1) = P(1' \text{ is transmitted}) = 0.4.$$

Also, given that the probability that a particular symbol is distorted is 0.01; that is,

$$P(1|0) = P(1 \text{ is received} | 0 \text{ is transmitted})$$

$$= 0.01 = P(0|1) = P(0 \text{ is received} | 1 \text{ is transmitted}).$$

Hence, from the total probability rule, the probability of receiving a zero is

$$P(1) = P(\text{received a 1}) = P(1|0)P(0) + P(1|1)P(1)$$

$$= (0.01)(0.6) + (0.99)(0.4) = 0.402.$$

Hence, irrespective of whether a 0 or 1 is transmitted, the probability of receiving a 1 is 0.402.

Bayes'e rule

Theorem 2.4.2: Assume $S = A_1 \cup A_2 \cup \dots \cup A_n$, where $P(A_i) > 0, i = 1, 2, \dots, n$ and $A_i \cap A_j = \emptyset$ for $i \neq j$. Then for any event B , with $P(B) > 0$

$$P(A_j | B) = \frac{P(A_j) P(B | A_j)}{\sum_{i=1}^n P(A_i) P(B | A_i)}.$$

Proof. We have

$$\begin{aligned} P(A_j | B) &= \frac{P(A_j \cap B)}{P(B)} \\ &= \frac{P(A_j \cap B)}{\sum_{i=1}^n P(A_i) P(B | A_i)}, \text{ by total probability rule for } P(B) \\ &= \frac{P(A_j) P(B | A_j)}{\sum_{i=1}^n P(A_i) P(B | A_i)}. \end{aligned}$$

Bayes' rule

- The probabilities $P(A_i)$ are called the **prior or a priori probabilities** of the events A_i .
- The conditional probability $P(A_j|B)$ is called the **posterior probability** of the event A_j .
- The events $A_1, \dots, A_n$ are sometimes called the states of nature.
- **Steps to apply Bayes's rule** (To find $P(A_1|D)$):
 - List all the probabilities including conditional probabilities given in the problem. That is $P(A_1), \dots, P(A_n)$ and $P(D|A_1), \dots, P(D|A_n)$.
 - Write the numerator as the product, $P(A_1)P(D|A_1)$.
 - Using total probability rule, find the denominator probability by calculating $P(D) = \sum_{i=1}^n P(A_i)P(D|A_i)$, in the Bayes' rule.
 - The desired probability is $\frac{\text{Numerator}}{\text{Denominator}}$.

Example 2.4.8 Suppose that three types of antimissile defense systems are being tested. From the design point of view, each of these systems has an equally likely chance of detecting and destroying an incoming missile within a range of 250 miles with a speed ranging up to nine times the speed of sound. However, in actual practice it has been observed that the precisions of these antimissile systems are not the same; that is

- the first system will usually detect and destroy the target 10 of 12 times
- the second will detect and destroy it 9 of 12 times
- the third will detect and destroy it 8 of 12 times

We have observed that a target has been detected and destroyed. What is the probability that the antimissile defense system was of the third type?

Bayes' rule

Solution

Let S_1 , S_2 , and S_3 be the events that the first, second, and third antimissile defense systems, respectively, are used. Also let D be the event that the target has been detected and destroyed. We wish to find $P(S_3|D)$. Given that $P(S_1) = P(S_2) = P(S_3) = 1/3$, $P(D|S_1) = 10/12$, $P(D|S_2) = 9/12$, and $P(D|S_3) = 8/12$. By total probability rule,

$$\begin{aligned} P(D) &= P(S_1)P(D|S_1) + P(S_2)P(D|S_2) + P(S_3)P(D|S_3) \\ &= \left(\frac{1}{3}\right)\left(\frac{10}{12}\right) + \left(\frac{1}{3}\right)\left(\frac{9}{12}\right) + \left(\frac{1}{3}\right)\left(\frac{8}{12}\right) = 0.75. \end{aligned}$$

Now using the Bayes' formula, we have

$$P(S_3|D) = \frac{P(S_3)P(D|S_3)}{P(D)} = \frac{(1/3)(8/12)}{0.75} = \frac{8}{27} = 0.2963.$$

If the target is destroyed, then the probability that the antimissile defense system was of the third type is 0.2963.

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions**
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Random variables

Definition 2.5.1 A random variable (r.v.) X is a function defined on a sample space, S , that associates a real number, $X(\omega) = x$, with each outcome ω in S .

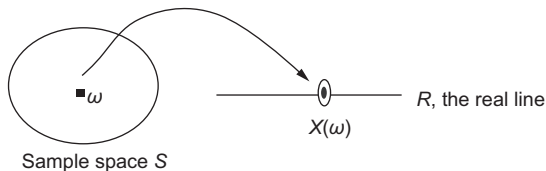


FIGURE 2.4 Random variable as a function.

Example 2.5.1 Two balanced coins are tossed and the face values are noted. Then the sample space $S = \{(H, H), (H, T), (T, H), (T, T)\}$. Define the random variable $X(\omega) = x$, where n is the number of heads and ω represents a simple event such as (H, H) .

EXAMPLE 2.5.1

Two balanced coins are tossed and the face values are noted. Then the sample space $S = \{(H,H), (H,T), (T,H), (T,T)\}$. Define the random variable $X(\omega) = n$, where n is the number of heads and ω represents a simple event such as (H,H) . Then

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = (T, T) \\ 1, & \text{if } \omega \in \{(H, T), (T, H)\} \\ 2, & \text{if } \omega = (H, H). \end{cases}$$

It can be noted that $X(\omega) = 0$ or 2 with probability $1/4$ (w.p. $1/4$) and $X(\omega) = 1$ w.p. $1/2$.

- The random variable is represented by a capital letter ($X, Y, Z, \dots$);
- Any particular real value of the random variable is denoted by the corresponding lowercase letter ($x, y, z, \dots$).
- Two types of random variables
 - discrete random variables
 - continuous random variables

Discrete random variables

Definition 2.5.2. A random variable X is said to be **discrete** if it can assume only a finite or countably infinite number of distinct values.

- We have seen that a discrete random variable assumes a finite or a countably infinite value.
- The discrete **probability mass function (pmf)** of a discrete random variable X is the function

$$p(x_i) = P(X = x_i), i = 1, 2, 3, \dots$$

- The probability function $p(x)$ is non-negative.
- As X must take on one of the values in $x_1, x_2, x_3, \dots$, we have

$$\sum_{i=1}^{\infty} p(x_i) = 1.$$

Discrete random variable

- The **cumulative distribution function (cdf)** F of the discrete random variable X is defined by

$$F(x) = P(X \leq x) = \sum_{all\ y \leq x} p(y), -\infty < x < \infty.$$

- It is also called a probability distribution function or simply the distribution function.
- Although the pmf $p(x)$ is defined only for a set of discrete values $x_1, x_2, x_3, \dots$, the cdf $F(x)$ is defined for all real numbers.

Example 2.5.3 Suppose that a fair coin is tossed twice so that the sample space is $S = \{(H, H), (H, T), (T, H), (T, T)\}$. Let X be number of heads. (a) Find the probability function for X . (b) Find the cumulative distribution function of X .

Discrete random variable

Solution

(a) We have

$$P(\{H, H\}) = P(\{H, T\}) = P(\{T, H\}) = P(\{T, T\}) = 1/4.$$

Hence, the pmf is given by

$$p(0) = P(X = 0) = 1/4, p(1) = 1/2, p(2) = 1/4.$$

(b) For example,

$$\begin{aligned} F(1.5) &= P(X \leq 1.5) = P(X = 0 \text{ or } 1) \\ &= P(X = 0) + P(X = 1) \\ &= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}. \end{aligned}$$

Proceeding similarly, we obtain (as shown in [Fig 2.5](#))

$$F(x) = \begin{cases} 0, & -\infty < x < 0 \\ 1/4, & 0 \leq x < 1 \\ 3/4, & 1 \leq x < 2 \\ 1, & 2 \leq x < \infty. \end{cases}$$

Continuous random variable

Let X be a random variable. Suppose that there exists a non-negative real-valued function: $f : \mathcal{R} \rightarrow [0, \infty)$ such that for any interval $[a, b]$,

$$P(X \in [a, b]) = \int_a^b f(t)dt.$$

Then X is called a **continuous** random variable.

- The function f is called the **probability density function (pdf)** of X .
- The **cumulative distribution function (cdf)** is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt.$$

Continuous random variable

- For a given function f to be a pdf, it needs to satisfy the following two conditions:
 - $f(x) \geq 0$ for all values of x
 - $\int_{-\infty}^{\infty} f(x)dx = 1$
- If f is continuous, we can get

$$\frac{dF(x)}{d(x)} = f(x)$$

- If f is the pdf of a random variable X , we can get

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Sketch map

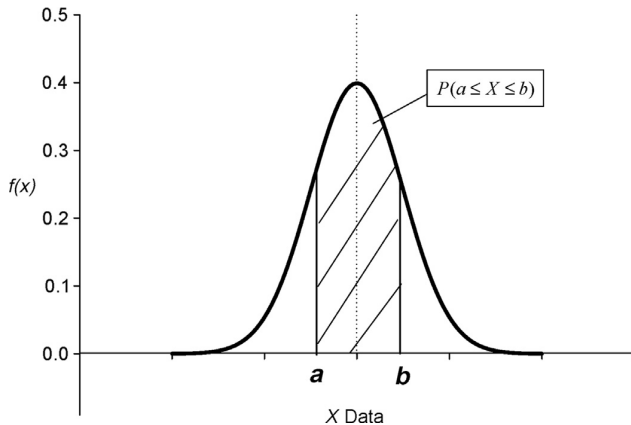


FIGURE 2.6 Probability as an area under a curve.

Distribution function

Some properties of distribution function:

- $0 \leq F(x) \leq 1$.
- $\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1$
- F is a nondecreasing function, and right continuous.

Example 2.5.4 Let the function

$$f(x) = \begin{cases} \lambda x e^{-x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) For what value of λ is f a pdf?
- (b) Find $F(x)$.

Distribution function

Solution

(a) First note that $f(x) \geq 0$. Now, for $f(x)$ to be a pdf, we need $\int_{-\infty}^{\infty} f(x)dx = 1$. Because $f(x) = 0$ for $x \leq 0$, therefore $\lambda = 1$. See [Fig. 2.7](#).

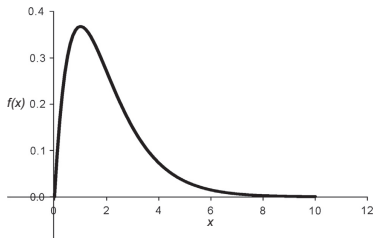


FIGURE 2.7 Graph of $f(x) = xe^{-x}$.

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f(x)dx = \int_0^{\infty} \lambda x e^{-x} dx \\ &= \lambda \int_{-\infty}^{\infty} x e^{-x} dx = \lambda \left[-x e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} dx \right] \text{ (using integration by parts)} \\ &= \lambda [0 - e^{-x} \Big|_0^{\infty}] = \lambda. \end{aligned}$$

Distribution function

(b) The cumulative distribution function is

$$F(x) = \int_{-\infty}^x f(t)dt = \begin{cases} 0, & x < 0 \\ \int_0^x te^{-te} dt = 1 - (x+1)e^{-x}, & x \geq 0. \end{cases}$$

Fig. 2.8 represents the cumulative distribution.

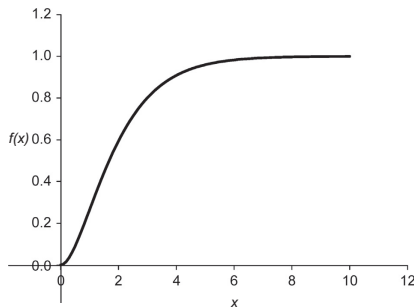


FIGURE 2.8 Graph of $F(x)$, $x \geq 0$.

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions**
- 7 Chapter summary
- 8 Computer examples and projects

Expected value

- Discrete case [Definition 2.6.2](#)

Let X be a discrete random variable with pmf $p(x)$. Then the expected value of X , denoted by $E(X)$, is defined by

$$\mu = E(X) = \sum_{\text{all } x} xp(x), \text{ provided } \sum_{\text{all } x} |x|p(x) < \infty$$

- Continuous case [Definition 2.6.3](#)

The expected value of a continuous random variable X with pdf $f(x)$ is defined by

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx, \text{ provided } \int_{-\infty}^{\infty} |x|f(x)dx < \infty$$

Expected value

- The expected value may be viewed as the balance point of the probability distribution on the real line, or in common language, the average
- The expected value of X is also called the expectation or mathematical expectation of X .
- We denote the expected value of X by μ .

Example 2.6.1 Let

$$X = \begin{cases} 1, & \text{with a probability } 1/2 \\ 0, & \text{with a probability } 1/2. \end{cases}$$

Find $E(X)$.

Solution $E(X) = 1(1/2) + 0(1/2) = 1/2$.

Expected value

EXAMPLE 2.6.2

Let X be a discrete random variable whose probability mass function is given in the following table:

x	-1	0	1	2	3	4	5
$P(x)$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{14}$	$\frac{2}{7}$	$\frac{1}{14}$	$\frac{1}{7}$	$\frac{1}{7}$

Find $E(X)$.

Solution

By definition,

$$\begin{aligned} E(X) &= \sum xp(x) = -1\left(\frac{1}{7}\right) + 0\left(\frac{1}{7}\right) + 1\left(\frac{1}{14}\right) \\ &\quad + 2\left(\frac{2}{7}\right) + 3\left(\frac{1}{14}\right) + 4\left(\frac{1}{7}\right) + 5\left(\frac{1}{7}\right) = 2. \end{aligned}$$

Expected value

EXAMPLE 2.6.3

Let $X \geq 0$ be an integer-valued random variable such that $P(X = n) = p_n$. Show that. $E(X) = \sum_{n=1}^{\infty} P(X \geq n)$.

Solution

Using the definition of expectation, and the fact that $(0)p_0 = 0$, we have

$$\begin{aligned} E(X) &= \sum_{n=1}^{\infty} np_n = 1p_1 + 2p_2 + 3p_3 + \cdots \\ &= p_1 + p_2 + p_3 + \cdots \\ &\quad + p_2 + p_3 + p_4 + \cdots \\ &\quad + p_3 + p_4 + \cdots \\ &= P(X \geq 1) + P(X \geq 2) + \cdots \\ &= \sum_{n=1}^{\infty} P(X \geq n) \end{aligned}$$

Definition 2.6.3 The **variance** of a random variable X is defined by

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2]$$

- The square root of variance, denoted by σ , is called the **standard deviation**.
- The variance is a measure of spread or variability of values of a random variable around the mean.

Expectation of function

Theorem 2.6.1 Let $g(X)$ be a function of X , if the sum or the integral exists the expected value of $g(X)$ is

$$E[g(X)] = \begin{cases} \sum_x g(x)p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x)f(x)dx, & \text{if } X \text{ is continuous} \end{cases}$$

Let c be a constant and let $g(X), g_1(X), \dots, g_n(X)$ be functions of a random variable X . Then the following results hold:

- $E(c) = c$
- $E[cg(X)] = cE[g(X)]$
- $E\left[\sum_i g_i(X)\right] = \sum_i E[g_i(X)]$
- $Var(aX + b) = a^2 Var(X)$. In particular, $Var(aX) = a^2 Var(X)$
- $Var(X) = E(X^2) - \mu^2$

Expectation of function

EXAMPLE 2.6.5

A discrete random variable X is said to be *uniformly distributed* over the numbers $1, 2, 3, \dots, n$, if

$$P(X = i) = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

Find $E(X)$ and $\text{Var}(X)$.

Solution

By definition

$$\begin{aligned} E(X) &= \sum_{i=1}^n x_i p_i \\ &= 1\left(\frac{1}{n}\right) + 2\left(\frac{1}{n}\right) + \dots + n\left(\frac{1}{n}\right) \\ &= \frac{1}{n} \left[\frac{n(n+1)}{2} \right] = \frac{n+1}{2}. \end{aligned}$$

Expectation of function

Similarly, using the summation formula $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$, we get

$$\begin{aligned} E(X^2) &= 1^2 \left(\frac{1}{n}\right) + 2^2 \left(\frac{1}{n}\right) + \cdots + n^2 \left(\frac{1}{n}\right) \\ &= \frac{1}{n} \left[\frac{n(n+1)(2n+1)}{6} \right] \\ &= \frac{(n+1)(2n+1)}{6}. \end{aligned}$$

Hence,

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (EX)^2 \\ &= \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{n^2 - 1}{12}. \end{aligned}$$

Expectation of function

EXAMPLE 2.6.6

To find out the prevalence of smallpox vaccine use, a researcher inquired into the number of times a randomly selected 200 people aged 16 and over in an African village had been vaccinated. He obtained the following figures: never, 17 people; once, 30; twice, 58; three times, 51; four times, 38; five times, 7. Assuming these proportions continue to hold exhaustively for the population of that village, what is the expected number of times those people in the village had been vaccinated, and what is the standard deviation?

Solution

Let X denote the random variable representing the number of times a person aged 16 or older in this village has been vaccinated. Then, we can obtain the following distribution:

x	0	1	2	3	4	5
$p(x)$	17/200	30/200	58/200	51/200	38/200	7/200

Expectation of function

Then,

$$\begin{aligned} E(X) &= \sum xp(x) = \frac{1}{200}(0(17) + 1(30) + 2(58) + 3(51) + 4(38) + 5(7)) \\ &= 2.43. \end{aligned}$$

Also,

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \sum x^2p(x) - (2.43)^2 = 7.52 - (2.43)^2 \\ &= 1.6151. \end{aligned}$$

Thus, the standard deviation is $\sqrt{1.6151} = 1.2709$.

Example 2.6.7 Let Y be a random variable with pdf

$$f(y) = \begin{cases} \frac{3}{64}y^2(4-y), & 0 \leq y \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

- Find the expected value and variance of Y .
- Let $X = 300Y + 50$. Find $E(X)$ and $Var(X)$.
- Find $P(X > 750)$.

Expectation of function

Solution

$$\begin{aligned}\text{(a)} \quad E(Y) &= \int_{-\infty}^{\infty} yf(y)dy \\ &= \frac{3}{64} \int_0^4 y^2(4-y)dy \\ &= 2.4\end{aligned}$$

and

$$\begin{aligned}Var(Y) &= \int_0^4 (y-2.4)^2 \frac{3}{64} y^2(4-y)dy \\ &= 0.64.\end{aligned}$$

Expectation of function

(b) Using the fact that $\text{Var}(aY + b) = a^2\text{Var}(Y)$, we have

$$\begin{aligned}\text{Var}(X) &= (300)^2 \text{Var}(Y) \\ &= 90,000(0.64) = 57,600. \\ P(X > 750) &= P(300Y + 50 > 750) \\ &= P\left(Y > \frac{7}{3}\right) \\ &= \frac{3}{64} \int_{7/3}^4 y^2(4 - y)dy = 0.55339.\end{aligned}$$

Moment

The k th moment about the origin of a random variable X

- EX^k
- μ'_k

The k th moment about its mean

- $E[(X - \mu)^k]$
- $\mu_k, k = 2, 3, 4, \dots$
- It is also called central k th moment of a random variable X

Some points

- $E(X) = \mu'_1 = \mu;$
- $\sigma^2 = \mu_2.$
- The second moment about mean (variance, σ^2) is used as a measure of dispersion about the mean.

The **skewness** of the distribution of X :

- It is the standardized third moment about mean

$$\alpha_3 = \frac{E(X - \mu)^3}{\sigma^3} = \frac{\mu_3}{\mu_2^{3/2}}$$

- Skewness is used as a measure of the asymmetry (lack of symmetry) of a density function about its mean.
- For symmetric distribution, $\alpha_3 = 0$.
- If $\alpha_3 = 0$, then we cannot say that the distribution is symmetric about the mean.

The **kurtosis** of the distribution:

- It is the standardized fourth moment about mean

$$\alpha_4 = \frac{E(X - \mu)^4}{\sigma^4}$$

- Kurtosis is a measure of whether the distribution is peaked or flat relative to a normal distribution.
- Distributions with relatively large tails are called **leptokurtic**, and those with small tails are called **platokurtic**.
- A distribution that has the same kurtosis as a normal distribution is known as **mesokurtic**.
- For a standard normal distribution, $\alpha_4 = 3$.

Moment-generating function

For a random variable X , suppose that there is a positive number h such that for $-h < t < h$ the mathematical expectation $E(e^{tX})$ exists. The moment-generating function (mgf) of the random variable X is defined by

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum e^{tx} p(x), & \text{if discrete} \\ \int e^{tx} f(x) dx, & \text{if continuous} \end{cases}$$

- An advantage of the **moment-generating function** is its ability to give the moments.

Moment-generating function

Recall that the Maclaurin series of the function e^{tx} is

$$e^{tx} = 1 + tx + \frac{(tx)^2}{2!} + \frac{(tx)^3}{3!} + \dots + \frac{(tx)^n}{n!} + \dots$$

By using the fact that the expected value of the sum equals the sum of the expected values, the moment-generating function can be written as

$$\begin{aligned} M_X(t) &= E[e^{tX}] = E\left[1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots + \frac{(tX)^n}{n!} + \dots\right] \\ &= 1 + tE[X] + \frac{t^2}{2!}E[X^2] + \frac{t^3}{3!}E[X^3] + \dots + \frac{t^n}{n!}E[X^n] + \dots \end{aligned}$$

Moment-generating function

Note that $M_X(0) = 1$ for all the distributions. Taking the derivative of $M_X(t)$ with respect to t , we obtain

$$\begin{aligned}\frac{dM_X(t)}{dt} = M'_X(t) &= E[X] + tE[X] + \frac{t^2}{2!}E[X^2] \\ &+ \frac{t^3}{3!}E[X^3] + \dots + \frac{t^{(n-1)}}{(n-1)!}E[X^n] + \dots\end{aligned}$$

Evaluating this derivative at $t = 0$, all terms except $E[X]$ become zero. We have

$$M'_X(0) = E[X].$$

Moment-generating function

Similarly, taking the second derivative of $M_X(t)$, we obtain

$$M_X''(0) = E[X^2].$$

Continuing in this manner, from the n th derivative $M_X^{(n)}(t)$ with respect to t , we obtain all the moments to be

$$M_X^{(n)}(0) = E[X^n], n = 1, 2, 3, \dots$$

- **Theorem 2.6.3** If $M_X(t)$ exists, then for any positive integer k ,

$$\left. \frac{d^k M_X(t)}{dt^k} \right|_{t=0} = M_X^{(k)}(0) = \mu'_k$$

- The usefulness of the theorem lies in the fact that, if the mgf can be found, the often difficult process of integration or summation involved in calculating different moments can be replaced by the much easier process of differentiation.

Moment-generating function

Properties of the moment-generating function:

- The moment-generating function of X is unique in the sense that, if two random variables X and Y have the same mgf ($M_X(t) = M_Y(t)$, for t in an interval containing 0), then X and Y have the same distribution.
- If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

That is, the mgf of the sum of two independent random variables is the product of the mgfs of the individual random variables. The result can be extended to n random variables.

- Let $Y = aX + b$, then $M_Y(t) = e^{bt}M_X(at)$.

Moment-generating function

EXAMPLE 2.6.8

Let X be a random variable with pf

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

(This random variable is called a binomial random variable, and the pmf is called a binomial distribution.) Show that $M_X(t) = [(1-p) + pe^t]^n$, for all real values of t . Also obtain mean and variance of the random variable X .

Solution

The moment-generating function of X is

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x}. \end{aligned}$$

Using the binomial formula, we have

$$M_X(t) = [pe^t + (1-p)]^n, \quad -\infty < t < \infty.$$

Moment-generating function

The first two derivatives of $M_X(t)$ are

$$M'_X(t) = n[(1-p) + pe^t]^{(n-1)}(pe^t)$$

and

$$M''_X(t) = n(n-1)[(1-p) + pe^t]^{(n-2)}(pe^t)^2 + n[(1-p) + pe^t]^{(n-1)}(pe^t).$$

Thus,

$$\mu = E(X) = M'_X(0) = np$$

and

$$\begin{aligned}\sigma^2 &= E(X^2) - \mu^2 = M''(0) - (np)^2 \\ &= n(n-1)p^2 + np - (np)^2 = np(1-p).\end{aligned}$$

Moment-generating function

EXAMPLE 2.6.11

Let X be a random variable with pdf $f(x) = (1/\sqrt{2\pi})e^{-x^2/2}$, $-\infty < x < \infty$. (We call such random variable a standard normal random variable.) Find the mgf of X .

Solution

By the definition of mgf, we have

$$\begin{aligned} E(e^{tx}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{tx} e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x^2 - 2tx)} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x^2 - 2tx + t^2) + \frac{t^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x-t)^2 + \frac{t^2}{2}} dx \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x-t)^2} dx = e^{t^2/2}. \end{aligned}$$

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary**
- 8 Computer examples and projects

Summary

- The concepts of random events and probability.
- Counting techniques.
- The concept of conditional probability, independence, and Bayes' rule.
- Random variables and distribution functions, moments, and moment-generating functions of random variables.

Contents

- 1 Introduction
- 2 Random events and probability
- 3 Counting techniques and calculation of probabilities
- 4 The conditional probability, independence, and Bayes' rule
- 5 Random variables and probability distributions
- 6 Moments and moment-generating functions
- 7 Chapter summary
- 8 Computer examples and projects

Example

- [Example 2.8.1](#) Calculating Cumulative Probabilities.
- [Example 2.8.2](#) Expected Value

Projects for chapter 2

- 2A The birthday problem

- Find the probability that in a group of k people no two have the same probability
- Find the probability that in a group of k people no two have the same probability
- Assuming that each sequence is equally likely, show that

$$q = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}$$

- Write a computer program for calculating q for $k = 2$ to 50, and find the first k for which $P > 5$. This will give the least number of people we should ask to make it an even chance that at least two people will have the same birthday.

2B The Hardy-Weinberg law

- Assuming that the genetic contributions of the mother and father are independent and the matings are random, show that the respective probabilities for the first-generation offspring are

$$p_1 = (p + q/2)^2, q_1 = 2(r + q/2)(p + q/2), r_1 = (r + q/2)^2$$

Also find $P(A)$ and $P(a)$.

- The Englishman G. H. Hardy and the German W. Weinberg could show that the foregoing probabilities in a population stay constant for generations if certain conditions are fulfilled. This is known as the HardyWeinberg law. Under the conditions of part (a), using the induction argument, show that the HardyWeinberg law is satisfied, i.e., $p_n = p_1, q_n = q_1$, and $r_n = r_1$ for all $n \geq 1$. The consequences of the HardyWeinberg law are that (1) no evolutionary change occurs through the process of sexual reproduction itself, and (2) changes in allele and genotype frequencies can result only from additional forces on the gene pool of a species.

2C Some basic probability simulation

- In 1000 coin tosses, what is the probability of having the same side come up 10 times in a row?
- In 10 coin tosses, what is the probability of having a different side come up with each throw, that is, that you never get two tails or two heads in a row?
- Write codes to generate numbers for the lottery you are interested in.
- Think of a few other situations where simulation is appropriate.

- Exercises
 - Exercises 2.2.16, 2.21
 - Exercises 2.3.12, 2.3.16
 - Exercises 2.4.12, 2.4.23
 - Exercises 2.5.8
 - Exercises 2.6.6, 2.6.8, 2.6.16
- Projects
 - 2A The birthday problem